Calibration-free discrimination of unresolved sources with symmetric multiphoton measurements

Prathamesh Balasaheb Kadam,[1,*] Prajwal S. Gaikwad,[1] Shreyas Subhash Raut,[1] and Tejas Uttam Shitole[1]

[1] *Department of Artificial Intelligence and Data Science, AISSMS Institute of Information Technology, Pune, India*

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A single star and a close binary can be told apart at the quantum limit by sorting photons into Hermite-Gauss modes, but only if the sorter is aligned with the centroid of the source. We compute exact error exponents for this test when the pointing is unknown and must be learned from photons. Measure-first learners estimate the centroid by direct imaging of calibration or target photons and then sort modes. They retain a fraction of the known-pointing quantum Chernoff exponent that depends only on the calibration budget: 17%–20% without calibration, and 33%, 63% and 69% with 1, 100 and 1000 calibration photons per target photon, at every separation. The residual pointing error turns the zero-background test into a background-limited one, and no single-photon receiver that we optimized escapes this limit. Projecting k stored target photons onto the symmetric subspace of their transverse states instead has zero background for every pointing offset and every static mode crosstalk. It attains a fraction $(k-1)/k$ of the quantum limit without any calibration. Groups of four photons already outperform 1000 calibration photons per target photon, and the gain in error exponent is 1.6 to 5.7. With a single target photon, joint learning keeps a constant 18% advantage in excess risk over measure-first learning, through an interference port that is dark for every pointing. Invariance under translations alone gains at most 1% over the two-photon Hong-Ou-Mandel test.

I. INTRODUCTION

Telling a single star from a close binary below the Rayleigh limit is a hypothesis test. Its quantum limit is reached by sorting the light into Hermite-Gauss (HG) modes centered on the source [1–3]. The error exponent of direct imaging vanishes as the fourth power of the separation, while the quantum exponent vanishes only as its square. The origin of the advantage is simple: a single source sends every photon into the fundamental mode, so a photon in any other mode signals a second source.

The same zero-background structure makes the advantage fragile. Crosstalk or misalignment of the sorter leaks single-source photons into the signal ports. The simple test then fails, and the optimal test recovers the quantum exponent only logarithmically as the leak vanishes [4]. A fixed offset removes the support mismatch on which the ideal test relies [5]. Measurements tailored to a known offset reduce the loss but do not remove it [6]. Existing analyses assume that the centroid is known [7], or treat the offset as fixed. In practice the pointing has to be learned from photons, either from calibration exposures or from the target itself [8].

This makes the test a problem of learning from quantum data [9–11]. A measure-first learner detects each photon and processes the outcomes classically. A coherent learner can store photons and measure them jointly. For estimation, collective measurements that exploit permutation symmetry are robust to misalignment [12], and two-photon interference beats the Rayleigh limit [13, 14]. For discrimination with a learned pointing, neither the cost of calibration nor a calibration-free alternative has been quantified.

We compute exact error exponents for both kinds of learner; all predictions were pre-registered (Appendix E). There are three main results.

1. Measure-first learners stay background-limited for any finite calibration budget. They retain a separation-independent fraction of the known-pointing exponent, from 0.17–0.20 without calibration to 0.69 with 1000 calibration photons per target photon (Sec. III).

2. Symmetric projection of stored photons removes the calibration problem. It has zero background for any pointing and any static crosstalk, attains $(k-1)/k$ of the quantum limit, and gives a 1.6- to 5.7-fold larger exponent (Sec. IV).

3. With one target photon, joint learning keeps a constant 18% advantage through a dark interference port (Sec. V).

Section VI shows that invariance under translations alone gains at most 1% over the Hong-Ou-Mandel (HOM) test, so permutation symmetry is the right invariance. Section VII discusses implementations and limitations.

II. ONE-VERSUS-TWO TESTING WITH AN UNKNOWN POINTING

We work in one transverse dimension, with a Gaussian point-spread function of unit width. In the weak-source regime a photon from a point source at x is in

* prathamesh.kadam@aissmsioit.org

the state $|\psi_x\rangle$, with amplitude $\psi(y-x) \propto e^{-(y-x)^2/4}$. In the HG basis these are coherent states of an oscillator, $\langle \mathrm{HG}_n|\psi_x\rangle = e^{-a^2/2}a^n/\sqrt{n!}$ with $a = x/2$, and their overlaps are $\langle\psi_x|\psi_y\rangle = e^{-(x-y)^2/8}$.

Under H_0 every photon is in $\rho_0(\theta) = |\psi_\theta\rangle\langle\psi_\theta|$. Under H_1 it is in $\rho_1(\theta) = \frac{1}{2}(|\psi_{\theta+d/2}\rangle\langle\psi_{\theta+d/2}| + |\psi_{\theta-d/2}\rangle\langle\psi_{\theta-d/2}|)$, an equal-brightness binary of separation d with the same centroid. We write $a_0 = d/4$. The pointing offset θ is unknown and static. The learner receives N target photons from one hypothesis (equal priors) and, optionally, κN labeled calibration photons from a known single source at the same pointing. Performance is measured by the error exponent per target photon, $E = \lim_{N\to\infty} -N^{-1}\ln P_{\mathrm{err}}$.

If θ is known, the quantum Chernoff exponent [15, 16] for the pure ρ_0 is $\xi_Q = -\ln\langle\psi_\theta|\rho_1|\psi_\theta\rangle = a_0^2$ exactly. It is attained by the null test $\{|\psi_\theta\rangle\langle\psi_\theta|, 1-|\psi_\theta\rangle\langle\psi_\theta|\}$, which never clicks under H_0 and clicks with probability $1-e^{-a_0^2}$ under H_1. We report all exponents as fractions of a_0^2.

III. MEASURE-FIRST LEARNERS STAY BACKGROUND-LIMITED FOR ANY CALIBRATION BUDGET

A measure-first learner estimates the centroid and then measures the remaining target photons at the estimate. Direct imaging attains the quantum Fisher information for the centroid. If the κN calibration photons and a fraction f of the target photons are imaged, the residual offset $t = \theta - \hat{\theta}$ has the large-deviation rate $g(t) = (\kappa + f)t^2/2$ per target photon. The other $(1-f)N$ target photons enter a single-photon receiver centered at $\hat{\theta}$, with outcome probabilities $p_{cj}(t)$ under H_c. The Bayes error exponent is (Appendix A)

$$E(\kappa, f) = \min_{t_0,t_1} \max_{0\le s\le 1} \Big[s\, g(t_0) + (1-s)\, g(t_1) - (1-f)\ln \sum_j p_{0j}(t_0)^s\, p_{1j}(t_1)^{1-s} \Big], \quad (1)$$

which we maximize over f. Here t_0 and t_1 are the worst-case offsets under the two hypotheses. We evaluate Eq. (1) for two receivers:

- HG sorting, with outputs $\mathrm{HG}_0, \ldots, \mathrm{HG}_5$ and a rest port;
- a real orthogonal six-mode receiver, optimized numerically to maximize Eq. (1).

The optimized receiver mixes modes of opposite parity, so its worst case has to be searched over offsets of both signs.

Table I and Fig. 1 show the result.

- *No calibration.* HG sorting retains 17.4% of the quantum limit, spending $f^* = 0.41$ of the target photons on centroiding. The optimized receiver retains 20.3%, because it extracts the centroid and the binary signal from the same photons.
- *With calibration.* Centroiding on the target stops paying ($f^* = 0$ for $\kappa \ge 1$), and the optimized receiver coincides with HG sorting.
- *Slow growth.* The retained fraction $F(\kappa)$ is 33% at $\kappa = 1$, 52% at 10, 63% at 100 and 69% at 1000; an exploratory point gives 82% at $\kappa = 10^8$.

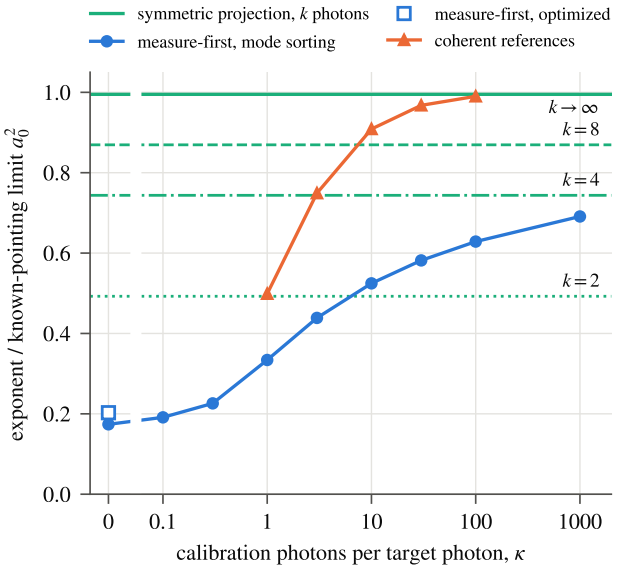


FIG. 1. Error exponent per target photon in units of the known-pointing limit a_0^2 ($a_0 = 0.1$). Circles: measure-first learning (direct-imaging centroiding, HG sorting) with κ calibration photons per target photon; $\kappa = 0$ is shown left of the axis break. Open square: optimized single-photon receiver at $\kappa = 0$. Triangles: each target photon projected jointly with κ calibration photons onto their symmetric subspace. Horizontal lines: symmetric projection of groups of k stored target photons, without calibration.

$F(\kappa)$ is the same for $a_0 = 0.05$, 0.1 and 0.2. The worst case pairs an offset single source with a centered binary, both Poissonian in the mode basis, so the exponent is homogeneous in a_0^2 (Appendix A).

The mechanism is a background.

- A residual offset t_0 sends single-source photons into HG_1 at the rate $u\,a_0^2$, with $u = t_0^2/4a_0^2$, while the binary sends photons there at the rate a_0^2.
- The Chernoff exponent between these two Poisson rates is $a_0^2 h(u)$, with $h(u) = \max_s[su + 1 - s - u^s]$. It recovers only as $1 - [\ln\ln(1/u) - 1]/\ln(1/u)$, the law found for crosstalk in Ref. [4].
- Balancing the cost $2(\kappa + f)u$ of a larger offset against this loss gives $E/a_0^2 \simeq \min_u[2(\kappa + f)u + (1-f)h(u)]$, which reproduces $F(\kappa)$.

TABLE I. Error exponents per target photon in units of the known-pointing quantum limit a_0^2 ($a_0 = 0.1$). Measure-first learners combine direct-imaging centroiding, with κ calibration photons per target photon and a fraction f^* of the target photons, with HG sorting or with an optimized six-mode receiver. "Coherent references" projects each target photon jointly with κ calibration photons onto their symmetric subspace. The measure-first values are identical to three digits for $a_0 = 0.05$ and 0.2. Generated from `results/qx3e/qx3e.json`.

κ	measure-first (SPADE)	f^*	measure-first (learned)	coherent references
0	0.174	0.41	0.203	–
0.1	0.191	0.29	–	–
0.3	0.226	0.17	–	–
1	0.334	0.00	0.334	0.499
3	0.439	0.00	–	0.749
10	0.525	0.00	0.525	0.909
30	0.582	0.00	–	0.968
100	0.629	0.00	0.629	0.990
1000	0.691	0.00	–	–
Symmetric projection of k stored target photons, no calibration:				
$k = 2$: 0.493, $k = 3$: 0.660, $k = 4$: 0.744, $k = 8$: 0.869, $k = 16$: 0.932; $k \to \infty$: 0.9950				

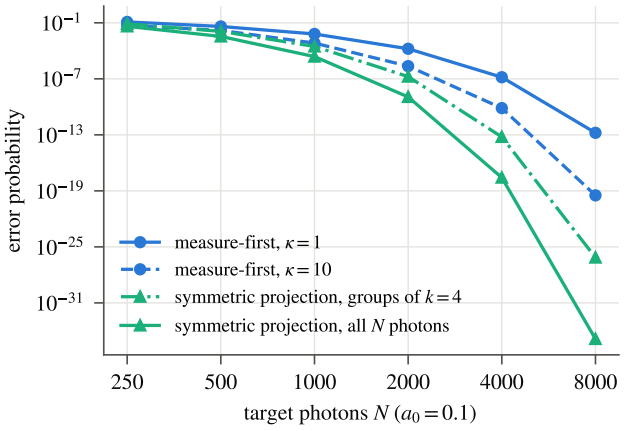


FIG. 2. Exact error probabilities versus the number of target photons N ($a_0 = 0.1$). Circles: measure-first learning with $\kappa = 1$ and 10. Triangles: symmetric projection in groups of $k = 4$ and of all N photons, $P_{\rm err} = \frac{1}{2} h_k^{N/k}$.

Every port that carries a second-order signal from the binary also carries a second-order background from the offset. Each target photon provides at most unit centroid Fisher information. No single-photon receiver therefore escapes the trade-off.

For finite photon numbers we computed the exact Bayes error of three-outcome HG sorting, summing the multinomial distribution over the normal residual offset. At $a_0 = 0.1$ and $N = 8000$ it is 1.6×10^{-13} for $\kappa = 1$ and 3.3×10^{-20} for $\kappa = 10$. The local exponents at $N = 4000$–8000 agree with Eq. (1) to within 3% (Fig. 2).

Takeaway. Learning the pointing and then measuring cannot recover the quantum advantage. Even 100 calibration photons per target photon leave the learner at 63% of the known-pointing exponent, at every separation.

IV. SYMMETRIC PROJECTION REMOVES THE CALIBRATION PROBLEM

A coherent learner can avoid learning θ. It stores k target photons in distinguishable time bins and projects their joint transverse state onto the symmetric subspace Sym^k.

- *Under* H_0 the k photons share the pure state $|\psi_\theta\rangle$, so the projection never fails. This holds for every θ and under any static crosstalk U, because $U^{\otimes k}$ preserves Sym^k.

- *Under* H_1 the photons are independent copies of $\rho_1(\theta)$, whose eigenvalues $\lambda_\pm = \frac{1}{2}(1 \pm e^{-2a_0^2})$ do not depend on θ. The projection succeeds with probability $h_k(\lambda_+, \lambda_-) = \sum_{j=0}^{k} \lambda_+^j \lambda_-^{k-j}$ (Appendix B).

Declaring two sources as soon as one group fails gives $P_{\rm err} = \frac{1}{2} h_k^{N/k}$, hence

$$E_k = -\tfrac{1}{k} \ln h_k(\lambda_+, \lambda_-) \simeq \tfrac{k-1}{k}\, a_0^2, \qquad (2)$$
$$E_\infty = -\ln \tfrac{1+e^{-2a_0^2}}{2} = a_0^2 - \tfrac{a_0^4}{2} + \ldots$$

For $k = 2$ this is the HOM test.

The exponents are $E_2 = 0.49$, $E_4 = 0.74$ and $E_8 = 0.87$, and $E_\infty = 0.9988$, 0.9950 and 0.9800 at $a_0 = 0.05$, 0.1 and 0.2, all in units of a_0^2 (Table I, Fig. 1).

- *Group sizes.* HOM pairs beat measure-first learning up to about three calibration photons per target photon, triples up to about 100, groups of four at 1000, and groups of eight even at the exploratory $\kappa = 10^8$.

- *Gain.* Large groups exceed the best measure-first exponent by a factor 4.9–5.7 without calibration, 3.0 at $\kappa = 1$, 1.9 at $\kappa = 10$ and 1.6 at $\kappa = 100$.

- *Finite N.* At $N = 8000$ and $a_0 = 0.1$ the error is about 10^{-35}, against 1.6×10^{-13} for measure-first learning with $\kappa = 1$.

Calibration photons can also be used coherently. Projecting each target photon together with κ fresh calibration photons fails under H_1 with probability $\kappa(1 - e^{-a_0^2})/(\kappa + 1)$ and never under H_0, so $E \simeq a_0^2\kappa/(\kappa + 1)$. For the same calibration budget this beats measuring the calibration photons by a factor 1.5 ($\kappa = 1$), 1.7 ($\kappa = 10$) and 1.6 ($\kappa = 100$).

Takeaway. A measurement that is blind to the pointing by symmetry recovers, without calibration, the advantage that measure-first learners lose to calibration. Four stored photons beat 1000 calibration photons per target photon.

V. WITH ONE TARGET PHOTON, JOINT LEARNING KEEPS A CONSTANT ADVANTAGE

The single-photon version of the problem connects to quantum learning theory [9–11].

- *Setting.* Let $\theta \sim \mathcal{N}(0, \tau^2)$. The learner receives n calibration photons and one target photon. Performance is the excess Bayes risk over the known-pointing error; the known-pointing advantage is $\frac{1}{2} - P_{\text{known}} = 0.678\, a_0^2$.

- *Exact reduction.* The calibration photons are unitarily equivalent to one collective coherent state $|\sqrt{n}\,\theta/2\rangle$ (Appendix C). This makes the joint Helstrom optimum exactly computable.

Results ($a_0 \in \{0.05, 0.1, 0.2\}$, $\tau \in \{0.2, 0.5\}$, $n \leq 128$; exploratory up to 1024):

- *Constant gap.* Measure-first learning (direct imaging of the calibration photons, then the Bayes test) has an excess risk 1.16–1.19 times that of the joint optimum for $n \geq 16$. The factor tends to 1.184 and is independent of a_0 to within 1% [Fig. 3(a)].

- *Plug-in is nearly optimal.* Estimating θ and applying the known-pointing Helstrom measurement costs at most 0.8% extra for $n \geq 16$.

- *Sign-blind calibration is useless.* HG counting of the calibration photons on the nominal axis recovers 95.6% of the known-pointing advantage at $n = 128$, against 95.5% with no calibration ($\tau = 0.2$). Its posterior is symmetric about the axis, so its average spread about the axis equals the prior spread [Fig. 3(b)].

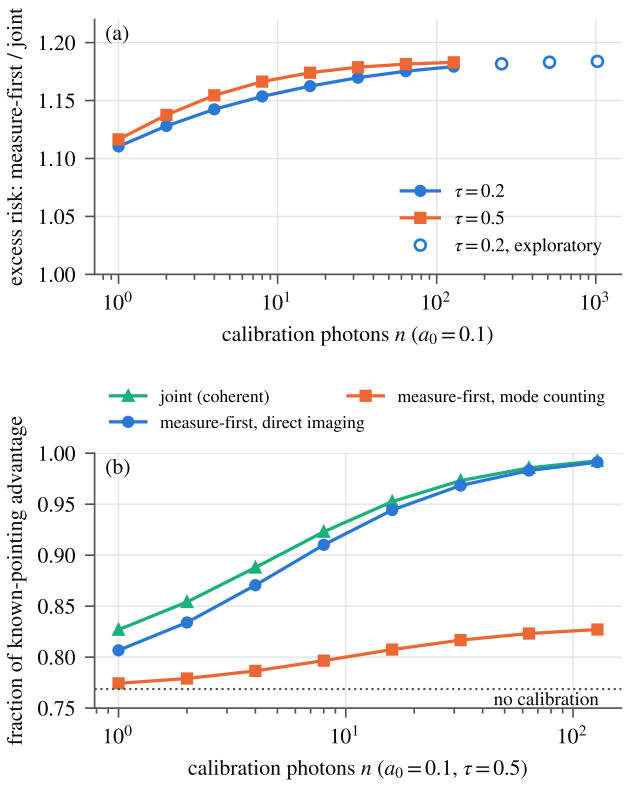


FIG. 3. One target photon. (a) Excess risk of measure-first learning divided by that of the joint optimum ($a_0 = 0.1$; $a_0 = 0.05$ and 0.2 agree within 1%). Open circles: exploratory, not pre-registered. (b) Fraction of the known-pointing advantage recovered ($\tau = 0.5$). Dotted: no calibration.

The mechanism is an interference port. Interfering the target photon with the collective calibration mode, with amplitude weights $\sqrt{n} : 1$, gives an output that is in vacuum under H_0 for every θ. Under H_1 that output carries a θ-independent binary with a_0^2 reduced by $n/(n+1)$.

- The joint learner therefore pays for the pointing uncertainty as a slightly smaller separation, with excess risk $\simeq 0.678\, a_0^2/(n+1)$. The dark port alone comes within 3% of the optimum at $n = 128$ ($\tau = 0.5$).

- The measure-first learner pays for it as an incoherent jitter, with excess risk $\simeq c \times 0.678\, a_0^2/(n+1/\tau^2)$, where $c \to 1.18$.

A constant asymptotic gap between joint and Gaussian measure-first learning is known for coherent-state templates [11]. Here it appears in the imaging problem, with a closed-form mechanism.

Takeaway. With one target photon the coherent advantage is a modest 18%. With many photons the same dark-port structure removes the background, and the advantage becomes decisive (Sec. IV).

VI. TRANSLATION INVARIANCE ALONE BUYS ALMOST NOTHING

Symmetric projection is invariant under every common mode transformation, whereas the physical nuisance is only a translation. A measurement invariant under translations alone might therefore extract more information. We tested this for two photons sharing a slow common jitter of width s_j and a static random crosstalk on the first 13 HG modes.

- *Only the relative coordinate is usable.* A translation-invariant measurement is block diagonal in the total momentum of the pair, and the total momentum is uninformative.
- *Its optimum.* The best such measurement refines HOM by projecting the relative coordinate onto its fundamental mode. It clicks under H_1 with probability $(1 - e^{-d^2/8})/2$, against $(1 - e^{-d^2/4})/4$ for HOM.
- *The gain.* The relative gain is $2/(1+e^{-d^2/8}) - 1 \simeq d^2/16$: 0.06%, 0.25% and 1.0% at $d = 0.1$, 0.2 and 0.4, independent of the jitter (Table II).
- *Fragility.* The coarse version of the test leaks $\approx 25\epsilon^2$ under H_0 for our crosstalk model, and falls below HOM already at $\epsilon = 10^{-4}$.
- *Learned measurements.* A two-photon measurement trained over a box of jitter and crosstalk values beats HOM by 0.2%–1.0% in the worst case; the best possible pair measurement beats HOM by 0.4%–1.5% there. The learned measurement is never below HOM, and is 1.5–1.7 times better than HOM at zero jitter.

Takeaway. Below the Rayleigh limit, full permutation symmetry costs almost nothing relative to the symmetry of the physical nuisance, and it buys immunity to crosstalk.

VII. DISCUSSION

Implementation.

- The symmetric projection requires k photons held in a quantum memory and released together. Only the photons within one group need to share the pointing, so drift between groups is harmless.
- For $k = 2$ it is the HOM test, a beam splitter followed by coincidence detection.
- For larger k it is weak Schur sampling of the transverse state. Implementations with atomic memories, multiport devices and linear optics have been proposed for estimation [12], and multiport interferometers resolve distinguishability classes of photons [17].
- Imperfect storage reintroduces a background. In a two-photon calculation with the same model (data in the repository), a random mode mismatch of strength ϵ between the two memories gives an H_0 background $\approx 3.8\epsilon^2$ for our mismatch model. A differential pointing jitter s_d between the stored photons gives $[1 - (1 + s_d^2)^{-1/2}]/2 \approx s_d^2/4$. Both must stay below the binary signal $\approx a_0^2$.

Limitations.

- One transverse dimension, an equal-brightness binary, a Gaussian point-spread function and ideal detectors.
- Asymptotic exponents, with a finite-N check at one separation.
- The measure-first family consists of direct-imaging centroiding followed by one fixed single-photon receiver. Adaptive multi-stage receivers are not excluded, although each target photon carries at most unit centroid information.

Unequal brightness (the exoplanet case), two-dimensional pointing and slow drift within a group remain open.

The broader lesson is that a measurement invariant under a nuisance replaces learning the nuisance. For sub-Rayleigh discrimination the right invariance is the permutation symmetry of stored photons. Its cost is a quantum memory, not more light.

ACKNOWLEDGMENTS

We thank the maintainers of NumPy, SciPy, PyTorch and Matplotlib, on which all numerical work in this paper depends. This work received no external funding.

Use of AI tools.

- *Tools.* This work made substantive use of Claude Opus 5.5 (Anthropic; model identifier `claude-opus-5-5`), run through Claude Code. Claude Sonnet 5.5 sub-agents were used for literature searches.
- *What it did, under the authors' direction:*
  - derived the analytic results;
  - wrote and ran the simulation code;
  - carried out the literature searches and novelty checks;
  - drafted the manuscript.
- *Direction and verification.* P. B. Kadam directed all of the work: he chose the research questions, approved each stage, and verified and checked all of the calculations reported here by hand.
- *Controls.*

TABLE II. Two-photon exponents relative to HOM. E_{T}: optimal translation-invariant measurement (any jitter). E_{pair}: optimal pair measurement at jitter s_j. ϵ^*: crosstalk at which the coarse translation-invariant test falls below HOM ($s_j = 0.4$). Generated from `results/qx2/qx2.json`.

d/σ	$E_{\mathrm{T}}/E_{\mathrm{HOM}}$	$E_{\mathrm{pair}}/E_{\mathrm{HOM}}$ ($s_j = 0$)	$E_{\mathrm{pair}}/E_{\mathrm{HOM}}$ ($s_j = 0.8$)	ϵ^*
0.1	1.0006	2.002	1.0009	1e-04
0.2	1.0025	2.008	1.0037	1e-04
0.4	1.0100	2.030	1.0149	1e-04
1	1.0643	2.197	1.0967	3e-03

– Every prediction was committed to a version-controlled pre-registration file before the code that tested it existed.

– All results were computed on the authors' hardware, and every figure and table is regenerated by scripts from version-controlled result files.

Data availability. The code and numerical results that support the findings of this article are openly available [18].

Appendix A: Large-deviation exponent of measure-first learners

Each imaged photon has a position distributed normally around θ with unit variance. With a flat prior on θ, the residual offset t is then normal with variance $[(\kappa+f)N]^{-1}$. (Under H_1 the true variance of target photons is $1 + d^2/4$, so this slightly favors the measure-first learner.) The remaining $M = (1-f)N$ target photons give counts $\boldsymbol{m}$, with probability

$$P_c(\boldsymbol{m}) = \int dt\, \phi_N(t)\, \mathrm{Mult}\big(\boldsymbol{m}; M, p_c(t)\big) \tag{A1}$$

under H_c. Write $x = \boldsymbol{m}/M$ for the empirical distribution. Sanov's theorem and Laplace's method give

$$-N^{-1} \ln P_c \to \min_t [g(t) + (1-f) D(x\|p_c(t))]. \tag{A2}$$

The Bayes error is dominated by the most likely confusable x, so $E = \min_{t_0,t_1} \min_x \max_c [g(t_c) + (1-f)D(x\|p_c(t_c))]$. Writing the maximum as a maximum over convex weights s, exchanging the minimum and maximum (Sion's theorem), and using $\min_x [sD(x\|p) + (1-s)D(x\|q)] = -\ln \sum_j p_j^s q_j^{1-s}$ gives Eq. (1).

Separation independence. Under HG sorting a single source at offset t_0 gives Poisson$(t_0^2/4)$ mode statistics. A centered binary gives Poisson(a_0^2), because both emitters have $a^2 = a_0^2$. For this pair the sum in Eq. (1) is $\exp[-st_0^2/4 - (1-s)a_0^2 + (t_0^2/4)^s a_0^{2(1-s)}]$, and $g(t_0) = 2(\kappa+f)a_0^2 u$ with $u = t_0^2/4a_0^2$. Every term is proportional to a_0^2, so E/a_0^2 depends on κ alone whenever $t_1 = 0$. The numerically found worst case has $|t_1| < 10^{-8} d$ at every κ and a_0.

Numerical details. Zero port probabilities restrict the sum to the common support, and the endpoints $s \to 0^+$ and 1^- are taken as one-sided limits. HG sorting satisfies $p_c(t) = p_c(-t)$, so offsets $t \geq 0$ suffice. Receivers that mix modes of opposite parity need offsets of both signs.

Appendix B: Symmetric-projection probabilities

Let $\rho = \sum_i \lambda_i |e_i\rangle\langle e_i|$. The symmetrized occupation-number states $|\boldsymbol{m}\rangle$ ($\sum_i m_i = k$) built from the e_i form an orthonormal basis of Sym^k, and $\rho^{\otimes k}$ acts on $|\boldsymbol{m}\rangle$ with eigenvalue $\prod_i \lambda_i^{m_i}$. Hence $\mathrm{Tr}[\Pi_{\mathrm{Sym}^k} \rho^{\otimes k}] = h_k(\lambda)$, the complete homogeneous symmetric polynomial.

- *Eigenvalues.* For $\rho_1 = \frac{1}{2}(|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta|)$ with $|\langle\alpha|\beta\rangle| = e^{-d^2/8} = e^{-2a_0^2}$, the eigenvalues are $\lambda_\pm = \frac{1}{2}(1 \pm e^{-2a_0^2})$. They are independent of θ and of any common unitary.
- *Invariance under H_0.* $(U|\psi_\theta\rangle)^{\otimes k} \in \mathrm{Sym}^k$ for every θ and U.
- *Exponent.* Expanding $h_k = \lambda_+^k \sum_{j=0}^k (\lambda_-/\lambda_+)^j$ gives $E_k \simeq (1-1/k)\lambda_- \simeq \frac{k-1}{k} a_0^2$. As $k \to \infty$, $E_k \to -\ln \lambda_+ < a_0^2$.

Projection with calibration photons. Write $|\phi\rangle = c|\psi\rangle + s|\psi^\perp\rangle$; the component $|\psi\rangle^{\otimes n}|\psi^\perp\rangle$ has squared norm $1/(n+1)$ after symmetrization. Hence

$$\mathrm{Tr}[\Pi_{\mathrm{Sym}^{n+1}}(|\psi\rangle\langle\psi|^{\otimes n} \otimes |\phi\rangle\langle\phi|)] = (1 + n|c|^2)/(n+1). \tag{B1}$$

With $|c|^2 = |\langle\psi_\theta|\psi_{\theta\pm d/2}\rangle|^2 = e^{-a_0^2}$, the projection fails under H_1 with probability $\kappa(1-e^{-a_0^2})/(\kappa+1)$.

Appendix C: Collective-mode reduction and the dark port

Reduction. A photon from a point source at x is in the coherent state $|x/2\rangle$ of the HG oscillator. The n-mode discrete Fourier transform of the photons' ladder operators maps $|\alpha\rangle^{\otimes n}$ to $|\sqrt{n}\,\alpha\rangle \otimes |0\rangle^{\otimes(n-1)}$, so the joint states are $\Omega_c = \int d\pi(\theta)\, |\sqrt{n}\theta/2\rangle\langle\sqrt{n}\theta/2| \otimes \tau_c(\theta)$. Here $\tau_0 = |\theta/2\rangle\langle\theta/2|$ and $\tau_1 = \frac{1}{2}\sum_\pm |\theta/2 \pm a_0\rangle\langle\theta/2 \pm a_0|$. This

unitary acts across the transverse degrees of freedom of different photons; it requires stored photons and is not a linear-optical operation on them.

Dark port. The mode $w = (B - \sqrt{n}\,b)/\sqrt{n+1}$, built from the collective mode B and the target mode b, has amplitude 0 under H_0 for every θ and $\mp a_0\sqrt{n/(n+1)}$ under H_1. It therefore carries a centered binary with $a_0^2 \to a_0^2 n/(n+1)$. Its excess risk is $\simeq 0.678\, a_0^2/(n+1)$, because the known-pointing advantage is $0.678\, a_0^2$.

Joint optimum.

- θ lies on a uniform grid over $\pm 6\tau$, with $K \geq 161$ points and $n\delta^2/8 \leq 0.01$.
- The calibration register is represented in the eigenbasis of $G_{ij} = \sqrt{w_i w_j}\, e^{-n(\theta_i-\theta_j)^2/8}$, keeping eigenvalues above 10^{-13} of the maximum (rank 7–87). The target photon uses HG_0–HG_{24}.
- $P = \frac{1}{2} - \frac{1}{4}\|\Omega_0 - \Omega_1\|_1$ follows from an eigendecomposition.
- Changing $K \to 2K-1$ changes the excess by at most 5×10^{-9} (relative), and changing the threshold to 10^{-11} by at most 10^{-12}.

Measure-first strategies.

- *Direct imaging.* The posterior is normal with variance $v = (n + 1/\tau^2)^{-1}$, and by translation invariance $P = \frac{1}{2} - \frac{1}{4}\|\sigma_0(v) - \sigma_1(v)\|_1$. Here $\sigma_c(v)$ is the target state averaged over an offset of variance v.
- *Plug-in.* It replaces the Bayes projector by the known-pointing Helstrom projector.
- *HG counting.* It uses the sufficient statistic $S \sim \mathrm{Poisson}(n\theta^2/4)$.

Appendix D: Numerical methods and checks

Measure-first exponents.

- Offsets lie on $\{0\} \cup 2d \times \{10^{-5}, \ldots, 1\}$ (121 log-spaced points), mirrored for non-even receivers. s uses 101 points with bounded refinement, followed by Nelder-Mead in (t_0, t_1).
- The optimized receivers are $W = e^{G-G^T}$ on HG_0–HG_5 plus a rest port, trained by Powell's method with three restarts (one from HG sorting).
- The $\kappa = 0$ optimum was rechecked on a dense two-sided grid and with 60 random starts.
- Finite-N errors use an 80-point Gauss-Hermite quadrature and the full multinomial sum; probability is conserved to 10^{-12}.

Two photons.

- HG pairs with $m+n \leq 24$, plus an exact remainder outcome; crosstalk acts on HG_0–HG_{12}.
- A 20-point Gauss-Hermite jitter rule.
- Pair bounds from coherent-state Gram matrices.
- Learned measurements are orthogonal maps on the 169-dimensional symmetric subspace, trained with Adam and then L-BFGS.

Faults fixed before any verdict. Faulty outputs are archived with the data.

1. A pair space $m+n \leq 18$, too small for crosstalk on 13 modes.
2. A one-sided offset search, which the optimized receivers exploited and so appeared 28%–35% better than HG sorting.
3. A memory crash with no output.

Software. Python 3.13.0, NumPy 2.2.4, SciPy 1.15.2, PyTorch 2.11.0. The AI assistance is described in the Acknowledgments.

Appendix E: Pre-registered predictions and outcomes

Table III lists every prediction behind Secs. III–VI, each committed before the code that tested it.

- *Refuted.* P4: plug-in is within 0.8% of Bayes, not 1.5–3 times worse, because misalignment enters the Helstrom projector only at second order.
- *Partly refuted.*
  - QE2: the fraction at $\kappa = 1$ is 0.334, just below the predicted range.
  - QE3: the optimized receiver gains 17% at $\kappa = 0$, against at most 10% predicted, because it centroids and detects with the same photons.
  - QE6: the finite-N exponent approaches its limit non-monotonically at $\kappa = 10$.

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TABLE III. Pre-registered predictions and outcomes. Predictions are quoted from the pre-registration addenda; outcomes are generated from the results files. E_{MF}, E_{COH}, E_{PLUG}, E_{SPADE}: excess risks of measure-first (direct imaging), joint, plug-in and HG-counting learners (Sec. V). E_{T}, E_{pair}, E_{HOM}: two-photon exponents (Sec. VI).

ID	Prediction (pre-registered)	Outcome	Verdict
Many target photons (Addendum I; Secs. III–IV)			
QE1	$E_\infty \geq 0.98\, a_0^2$; E_k within 3% of $(k-1)/k$	0.9988, 0.9950, 0.9800; max deviation 1.5%	confirmed
QE2	$F(\kappa)$ in $[0.15, 0.30]$, $[0.35, 0.50]$, $[0.50, 0.67]$, $[0.60, 0.75]$, < 0.85 at $\kappa = 0, 1, 10, 100, 1000$; a_0-variation $< 10\%$	0.174, 0.334, 0.525, 0.629, 0.691; variation 0.00%	partly refuted ($\kappa = 1$)
QE3	optimized receiver $\leq 1.1\times$ mode sorting	0.984, 1.000, 1.000, 1.000, 1.166, 1.000, 1.000, 1.000 ($a_0 = 0.05$, then 0.1; $\kappa = 0, 1, 10, 100$)	partly refuted ($\kappa = 0$)
QE4	$E_\infty/F \geq 3$ at $\kappa = 0$, ≥ 1.3 at $\kappa = 100$; HOM beats measure-first for $\kappa \leq 1$	5.72, 1.58; HOM/measure-first ≥ 1.47	confirmed
QE5	coherent references $\geq 1.1\times$ measure-first ($\kappa = 1$–100)	1.49–1.73	confirmed
QE6	finite-N local exponent approaches the large-deviation value monotonically, within 30% at $N = 8000$	$\kappa = 1$: 1.35, 1.11, 1.09, 1.05, 1.03; $\kappa = 10$: 0.93, 1.19, 1.08, 0.99, 1.03	partly refuted (monotonicity)
One target photon (Addendum H; Sec. V)			
P1	$E_{\mathrm{COH}} \leq E_{\mathrm{MF}} \leq E_{\mathrm{PLUG}}$, $E_{\mathrm{COH}} \leq E_{\mathrm{SPADE}}$	holds for all $n \geq 1$	confirmed
P2	$E_{\mathrm{MF}}/E_{\mathrm{COH}} \leq 1.2$ for $n \geq 16$	1.161–1.188	confirmed
P3	ratio varies $< 10\%$ across a_0 at $n = 64$	$< 1\%$	confirmed
P4	$E_{\mathrm{PLUG}}/E_{\mathrm{MF}} \in [1.5, 3]$ for $n \geq 16$	1.001–1.008	refuted
P5	$E_{\mathrm{SPADE}}/E_{\mathrm{MF}} \geq 5$ at $n = 128$, $\tau = 0.5$	19.5–19.8	confirmed
P6	$E_{\mathrm{MF}}(n + 1/\tau^2)$ constant within $\pm 15\%$ on $n \in [16, 128]$	within $-2.6\%/+1.8\%$	confirmed
P7	n_{90} independent of a_0	0 ($\tau = 0.2$), 8 ($\tau = 0.5$) for all a_0	confirmed
Two photons, invariance (Addendum H; Sec. VI)			
T1	$E_{\mathrm{T}}/E_{\mathrm{HOM}} \leq 1.011$ ($d \leq 0.4$), $\in [1.05, 1.08]$ ($d = 1$)	1.0006, 1.0025, 1.0100, 1.0643	confirmed
T2	$E_{\mathrm{pair}}/E_{\mathrm{T}} \leq 1.15$ at $s_j = 0.8$	max 1.0048	confirmed
T3	coarse test falls below HOM at $\epsilon^* \leq 0.01$	$\epsilon^* \leq 3 \times 10^{-4}$	confirmed
T4	refined test $\geq$ HOM everywhere; gain at $\epsilon = 0.1$ $\leq 0.2\times$ gain at $\epsilon = 0$	min difference ≥ 0; 4%–7%	confirmed
T5	learned pair measurement ≥ 0.99 of pair bound (nuisance known)	min 0.9994	confirmed
T6	learned minimax / HOM $\in [1.00, 1.10]$; ancillas add $\leq 2\%$	1.0023, 1.0097 ($d = 0.2, 0.4$)	confirmed

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